AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM2.5 (pm25)
Observed / expected baseline	42 / 12.8663 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-18 13:00 UTC
Location	29.7690, -95.3179
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 637	54.54 to 100; mean 82.9069	94.33 at 2026-06-18 12:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 637	23 to 35; mean 28.4875	28 at 2026-06-18 12:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 638	0 to 360; mean 134.53	190 at 2026-06-18 12:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 639	0 to 13.3755; mean 4.1083	2.57222 at 2026-06-18 12:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5850.83 to 5918.16; mean 5882.92	5868.11 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	17.322 to 1845; mean 630.985	417.381 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	38.0666 to 59.8281; mean 49.24	49.5931 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	995.556 to 1012.08; mean 1004.21	1003.57 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	17.6773 to 23.2907; mean 20.1303	19.7866 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-6.14388 to 1.58572; mean -1.6176	0.165723 at 2026-06-18 12:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	-1.37312 to 6.26864; mean 2.472	2.64051 at 2026-06-18 12:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	18 / 1173	0.001 to 0.049; mean 0.024	0.013 at 2026-06-18 13:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 215	9 to 72; mean 34.1814	68 at 2026-06-18 13:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	82 / 2565	0 to 60; mean 14.444	42 at 2026-06-18 13:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 48.5333	81 at 2026-06-18 13:00Z; at event time
openweather	Relative humidity (humidity)	percent	5 / 360	60 to 97; mean 84.0944	89 at 2026-06-18 13:00Z; at event time
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 8.38; mean 0.2468	0 at 2026-06-18 13:00Z; at event time
openweather	Surface pressure (pressure)	hPa	5 / 360	1001 to 1012; mean 1006.61	1006 at 2026-06-18 13:00Z; at event time
openweather	Air temperature (temperature)	degC	5 / 360	23.79 to 35.6; mean 28.8575	27.75 at 2026-06-18 13:00Z; at event time
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 360; mean 148.478	197 at 2026-06-18 13:00Z; at event time
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 6.85; mean 2.3778	2.24 at 2026-06-18 13:00Z; at event time
purpleair	PM2.5 (pm25)	ug/m3	70 / 4947	0 to 102.5; mean 15.5881	39.8 at 2026-06-18 13:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	5 / 5	0.0278085 to 0.0281594; mean 0.0279	0.0281594 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	9.81221e-05 to 0.000167838; mean 0.0001	9.81221e-05 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	2 / 2	4.17195e-05 to 5.0006e-05; mean 0	4.17195e-05 at 2026-06-17 20:11Z; 16 h 49 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	2 / 2	1 to 1; mean 1	1 at 2026-06-17 20:11Z; 16 h 49 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-4.97356e-05 to -4.29344e-06; mean -0	-4.29344e-06 at 2026-06-18 19:23Z; 6 h 24 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-18 19:23Z; 6 h 24 min after event
tceq	Carbon monoxide (co)	ppm	3 / 216	0.1 to 0.7; mean 0.2079	0.2 at 2026-06-18 13:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 851	-2.7 to 23.4; mean 4.5974	11.1 at 2026-06-18 13:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 216	-0.5 to 1.8; mean 0.0449	0.3 at 2026-06-18 13:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-17 00Z 91.17, 06Z 95.39, 12Z 86.13, 18Z 71.8, 06-18 00Z 86.58, 06Z 95.32, 12Z 76.47, 18Z 66.14, 06-19 00Z 86.37, 06Z 92.99, 12Z 74.29, 18Z 73.87, 06-20 00Z 83.93
asos	Air temperature (temperature)	06-17 00Z 25.96, 06Z 24.8, 12Z 26.2, 18Z 29.8, 06-18 00Z 26.68, 06Z 26.12, 12Z 30.79, 18Z 33.32, 06-19 00Z 28.99, 06Z 27.38, 12Z 30.71, 18Z 30.87, 06-20 00Z 27.13
asos	Wind direction (wind_direction)	06-17 00Z 108.4, 06Z 101.3, 12Z 118.1, 18Z 86.85, 06-18 00Z 85.56, 06Z 128, 12Z 188.1, 18Z 172.4, 06-19 00Z 153.8, 06Z 140.8, 12Z 164.1, 18Z 170.9, 06-20 00Z 92.5
asos	Wind speed (wind_speed)	06-17 00Z 2.713, 06Z 3.921, 12Z 7.251, 18Z 6.335, 06-18 00Z 2.496, 06Z 1.858, 12Z 4.077, 18Z 5.04, 06-19 00Z 4.086, 06Z 3.096, 12Z 4.592, 18Z 3.611, 06-20 00Z 4.051
noaa_gfs	500-hPa geopotential height (gh_500)	06-17 06Z 5858, 12Z 5852, 18Z 5855, 06-18 00Z 5861, 06Z 5873, 12Z 5869, 18Z 5894, 06-19 00Z 5901, 06Z 5918, 12Z 5905, 18Z 5910, 06-20 00Z 5899
noaa_gfs	Planetary boundary layer height (pbl_height)	06-17 06Z 137.1, 12Z 286.1, 18Z 953.1, 06-18 00Z 670, 06Z 103.1, 12Z 309.3, 18Z 1529, 06-19 00Z 785.9, 06Z 463.2, 12Z 551.4, 18Z 1593, 06-20 00Z 190
noaa_gfs	Precipitable water (precipitable_water)	06-17 06Z 56.04, 12Z 54.99, 18Z 54.97, 06-18 00Z 48.55, 06Z 45.1, 12Z 47.88, 18Z 44.75, 06-19 00Z 42.03, 06Z 40.38, 12Z 45.26, 18Z 51.92, 06-20 00Z 59.02
noaa_gfs	Surface pressure (surface_pressure)	06-17 06Z 1005, 12Z 1003, 18Z 1002, 06-18 00Z 999.1, 06Z 1001, 12Z 1002, 18Z 1003, 06-19 00Z 1003, 06Z 1006, 12Z 1008, 18Z 1009, 06-20 00Z 1008
noaa_gfs	850-hPa temperature (t_850)	06-17 06Z 18.05, 12Z 18.17, 18Z 18.44, 06-18 00Z 21.03, 06Z 21.08, 12Z 19.68, 18Z 19.44, 06-19 00Z 22.06, 06Z 22.86, 12Z 22.07, 18Z 18.8, 06-20 00Z 19.88
noaa_gfs	10-m eastward wind (u_10m)	06-17 06Z -2.335, 12Z -2.876, 18Z -5.036, 06-18 00Z -2.603, 06Z -0.4446, 12Z 0.1984, 18Z -0.2216, 06-19 00Z -1.502, 06Z -1.482, 12Z -1.092, 18Z -0.2103, 06-20 00Z -1.808
noaa_gfs	10-m northward wind (v_10m)	06-17 06Z 0.9349, 12Z 0.4399, 18Z 3.126, 06-18 00Z -0.7859, 06Z 1.633, 12Z 2.484, 18Z 4.365, 06-19 00Z 4.397, 06Z 2.912, 12Z 3.328, 18Z 4.465, 06-20 00Z 2.365
openaq	Ozone (ozone)	06-17 00Z 0.01346, 06Z 0.01935, 12Z 0.02179, 18Z 0.03299, 06-18 00Z 0.02502, 06Z 0.01554, 12Z 0.02761, 18Z 0.03682, 06-19 00Z 0.02806, 06Z 0.01769, 12Z 0.02035, 18Z 0.02654, 06-20 00Z 0.02259
openaq	PM10 (pm10)	06-17 00Z 19.8, 06Z 14.44, 12Z 21.5, 18Z 27.83, 06-18 00Z 24.61, 06Z 23.17, 12Z 51.11, 18Z 49.29, 06-19 00Z 44.67, 06Z 38.71, 12Z 48.44, 18Z 45.94, 06-20 00Z 25.67
openaq	PM2.5 (pm25)	06-17 00Z 2.71, 06Z 4.772, 12Z 5.696, 18Z 2.745, 06-18 00Z 4.291, 06Z 8.748, 12Z 30.04, 18Z 20.34, 06-19 00Z 33.7, 06Z 21.82, 12Z 14.5, 18Z 10.17, 06-20 00Z 6.734
openweather	Cloud cover (cloud_cover)	06-17 00Z 96.04, 06Z 75.69, 12Z 24.55, 18Z 70.83, 06-18 00Z 25.2, 06Z 3.133, 12Z 39.5, 18Z 19.4, 06-19 00Z 6.931, 06Z 80.77, 12Z 72.39, 18Z 68.6, 06-20 00Z 89.4
openweather	Relative humidity (humidity)	06-17 00Z 90.6, 06Z 93.86, 12Z 91.58, 18Z 75.87, 06-18 00Z 83.67, 06Z 93.03, 12Z 82.5, 18Z 70.27, 06-19 00Z 83.59, 06Z 91.3, 12Z 81.81, 18Z 72.73, 06-20 00Z 81.4
openweather	Precipitation rate (precipitation)	06-17 00Z 0.5164, 06Z 0.7976, 12Z 1.38, 18Z 0.079, 06-18 00Z 0.1477, 06Z 0, 12Z 0.009, 18Z 0, 06-19 00Z 0, 06Z 0, 12Z 0, 18Z 0.035, 06-20 00Z 0.378
openweather	Surface pressure (pressure)	06-17 00Z 1008, 06Z 1006, 12Z 1005, 18Z 1003, 06-18 00Z 1003, 06Z 1004, 12Z 1006, 18Z 1005, 06-19 00Z 1008, 06Z 1009, 12Z 1011, 18Z 1011, 06-20 00Z 1011
openweather	Air temperature (temperature)	06-17 00Z 26.14, 06Z 25.04, 12Z 25.53, 18Z 30.41, 06-18 00Z 27.38, 06Z 26.05, 12Z 30.23, 18Z 34.37, 06-19 00Z 29.97, 06Z 27.74, 12Z 30.18, 18Z 32.75, 06-20 00Z 28.92
openweather	Wind direction (wind_direction)	06-17 00Z 107, 06Z 111.2, 12Z 107.1, 18Z 115.9, 06-18 00Z 107.4, 06Z 197.5, 12Z 177.4, 18Z 178.3, 06-19 00Z 158.7, 06Z 165.5, 12Z 163.5, 18Z 185.2, 06-20 00Z 149.2
openweather	Wind speed (wind_speed)	06-17 00Z 2.362, 06Z 2.014, 12Z 3.329, 18Z 2.887, 06-18 00Z 1.624, 06Z 2.168, 12Z 2.35, 18Z 2.545, 06-19 00Z 2.314, 06Z 2.298, 12Z 2.28, 18Z 2.305, 06-20 00Z 2.446

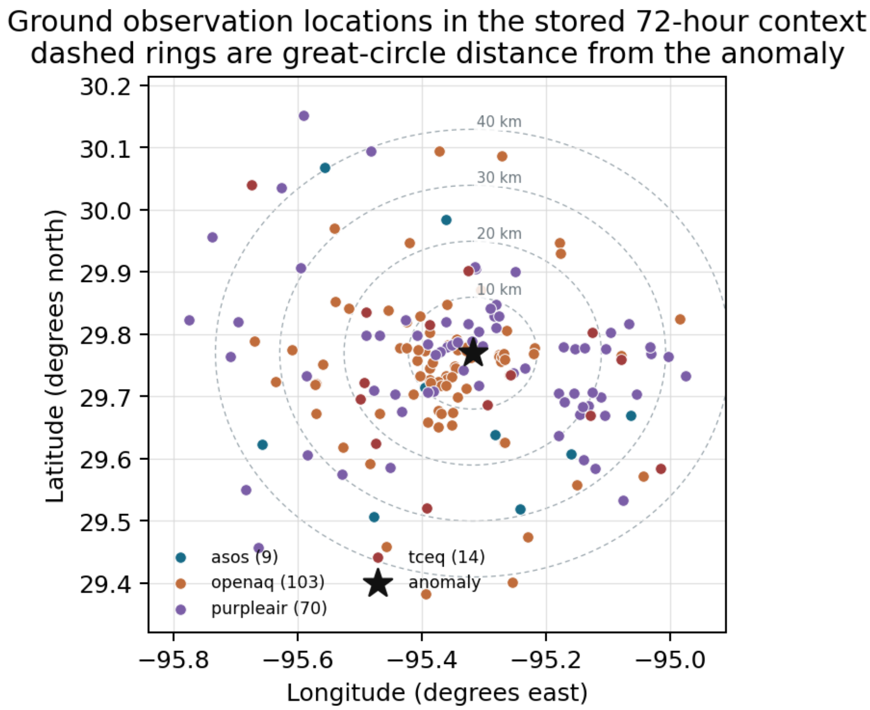
Source	Metric	Values
purpleair	PM2.5 (pm25)	06-17 00Z 5.053, 06Z 7.318, 12Z 8.16, 18Z 4.005, 06-18 00Z 5.901, 06Z 16.34, 12Z 30.86, 18Z 24.51, 06-19 00Z 34.18, 06Z 24.21, 12Z 15.54, 18Z 11.76, 06-20 00Z 7.323
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
sentinel5p	CO column (s5p_co_column)	06-17 18Z 0.02786, 06-18 18Z 0.02816, 06-19 18Z 0.02782
sentinel5p	CO granules available (s5p_co_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-17 18Z 0.0001631, 06-18 18Z 9.812e-05, 06-19 18Z 0.0001451
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-17 18Z 4.172e-05, 06-19 18Z 5.001e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-17 18Z 1, 06-19 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-17 18Z -1.182e-05, 06-18 18Z -4.293e-06, 06-19 18Z -2.9e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-17 18Z 1, 06-18 18Z 1, 06-19 18Z 1
tceq	Carbon monoxide (co)	06-17 00Z 0.2067, 06Z 0.1556, 12Z 0.1889, 18Z 0.1611, 06-18 00Z 0.1722, 06Z 0.2556, 12Z 0.2933, 18Z 0.2278, 06-19 00Z 0.1944, 06Z 0.1944, 12Z 0.2278, 18Z 0.2222, 06-20 00Z 0.2333
tceq	Nitrogen dioxide (no2)	06-17 00Z 6.797, 06Z 4.976, 12Z 4.426, 18Z 3.517, 06-18 00Z 4.718, 06Z 5.821, 12Z 4.287, 18Z 2.812, 06-19 00Z 3.226, 06Z 3.9, 12Z 4.366, 18Z 4.846, 06-20 00Z 7.939
tceq	Sulfur dioxide (so2)	06-17 00Z 0.02, 06Z -0.1, 12Z -0.08333, 18Z -0.04444, 06-18 00Z 0.01667, 06Z -0.05556, 12Z 0.1467, 18Z 0.1944, 06-19 00Z 0.4611, 06Z -0.05556, 12Z 0.06111, 18Z 0.005556, 06-20 00Z 0

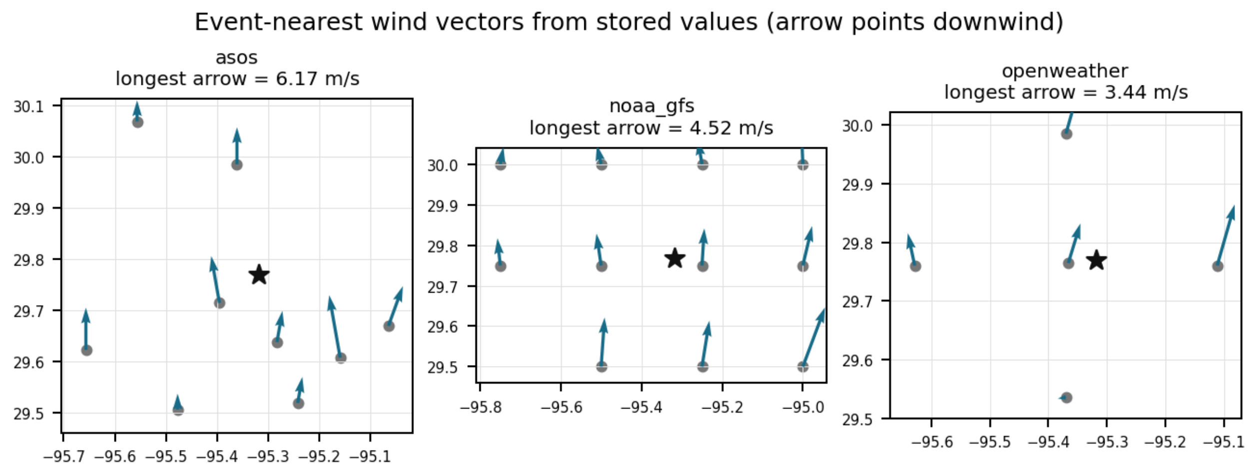
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (9.638 km) 85.03, HOU (15.02 km) 82.07, EFD (23.659 km) 83.79, IAH (24.303 km) 82.9, T41 (26.902 km) 84.19, LVJ (28.769 km) 82.05, AXH (33.03 km) 79.13, SGR (36.553 km) 83.77, +1 more
asos	Air temperature (temperature)	MCJ (9.638 km) 28.21, HOU (15.02 km) 28.83, EFD (23.659 km) 28.7, IAH (24.303 km) 28.36, T41 (26.902 km) 28.51, LVJ (28.769 km) 28.64, AXH (33.03 km) 28.68, SGR (36.553 km) 28.56, +1 more
asos	Wind direction (wind_direction)	MCJ (9.638 km) 144.4, HOU (15.02 km) 140, EFD (23.659 km) 127.5, IAH (24.303 km) 137.4, T41 (26.902 km) 145, LVJ (28.769 km) 135, AXH (33.03 km) 113.3, SGR (36.553 km) 142.6, +1 more
asos	Wind speed (wind_speed)	MCJ (9.638 km) 5.416, HOU (15.02 km) 4.48, EFD (23.659 km) 4.997, IAH (24.303 km) 4.016, T41 (26.902 km) 5.009, LVJ (28.769 km) 3.787, AXH (33.03 km) 2.408, SGR (36.553 km) 4.258, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (6.883 km) 5883, gfs:29.75,-95.50 (17.709 km) 5883, gfs:30.00,-95.25 (26.504 km) 5883, gfs:29.50,-95.25 (30.624 km) 5883, gfs:29.75,-95.00 (30.756 km) 5882, gfs:30.00,-95.50 (31.113 km) 5883, gfs:29.50,-95.50 (34.709 km) 5883, gfs:30.00,-95.00 (39.984 km) 5882, +3 more

Source	Metric	Values
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (6.883 km) 673.4, gfs:29.75,-95.50 (17.709 km) 736.2, gfs:30.00,-95.25 (26.504 km) 681.2, gfs:29.50,-95.25 (30.624 km) 563, gfs:29.75,-95.00 (30.756 km) 623.1, gfs:30.00,-95.50 (31.113 km) 768.3, gfs:29.50,-95.50 (34.709 km) 570.9, gfs:30.00,-95.00 (39.984 km) 501.5, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (6.883 km) 49.83, gfs:29.75,-95.50 (17.709 km) 48.95, gfs:30.00,-95.25 (26.504 km) 48.92, gfs:29.50,-95.25 (30.624 km) 50.03, gfs:29.75,-95.00 (30.756 km) 50.58, gfs:30.00,-95.50 (31.113 km) 48.14, gfs:29.50,-95.50 (34.709 km) 49.43, gfs:30.00,-95.00 (39.984 km) 49.85, +3 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (6.883 km) 1005, gfs:29.75,-95.50 (17.709 km) 1004, gfs:30.00,-95.25 (26.504 km) 1004, gfs:29.50,-95.25 (30.624 km) 1006, gfs:29.75,-95.00 (30.756 km) 1006, gfs:30.00,-95.50 (31.113 km) 1002, gfs:29.50,-95.50 (34.709 km) 1005, gfs:30.00,-95.00 (39.984 km) 1005, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (6.883 km) 20.08, gfs:29.75,-95.50 (17.709 km) 20.15, gfs:30.00,-95.25 (26.504 km) 20.01, gfs:29.50,-95.25 (30.624 km) 20.19, gfs:29.75,-95.00 (30.756 km) 20.11, gfs:30.00,-95.50 (31.113 km) 20.1, gfs:29.50,-95.50 (34.709 km) 20.18, gfs:30.00,-95.00 (39.984 km) 19.97, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (6.883 km) -1.392, gfs:29.75,-95.50 (17.709 km) -1.615, gfs:30.00,-95.25 (26.504 km) -1.623, gfs:29.50,-95.25 (30.624 km) -1.616, gfs:29.75,-95.00 (30.756 km) -1.562, gfs:30.00,-95.50 (31.113 km) -1.575, gfs:29.50,-95.50 (34.709 km) -1.783, gfs:30.00,-95.00 (39.984 km) -1.853, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (6.883 km) 2.264, gfs:29.75,-95.50 (17.709 km) 2.185, gfs:30.00,-95.25 (26.504 km) 1.898, gfs:29.50,-95.25 (30.624 km) 2.962, gfs:29.75,-95.00 (30.756 km) 3.084, gfs:30.00,-95.50 (31.113 km) 1.872, gfs:29.50,-95.50 (34.709 km) 2.463, gfs:30.00,-95.00 (39.984 km) 2.27, +3 more
openaq	Ozone (ozone)	3378 (7.015 km) 0.01988, 2046 (7.328 km) 0.01707, 1319333 (7.534 km) 0.02732, 2039 (9.388 km) 0.02191, 325 (9.456 km) 0.02193, 310 (14.708 km) 0.02678, 320 (16.681 km) 0.02123, 319 (18.051 km) 0.02189, +10 more
openaq	PM10 (pm10)	15852419 (0.0 km) 39.59, 13380082 (7.015 km) 34.96, 271 (21.343 km) 28.07
openaq	PM2.5 (pm25)	15852418 (0.0 km) 18.93, 15588191 (0.889 km) 13.9, 8967237 (1.063 km) 6.989, 12256170 (1.277 km) 14.65, 8967273 (1.528 km) 10.95, 8967217 (1.567 km) 12.71, 8967005 (1.748 km) 15.12, 15596557 (2.479 km) 14.53, +74 more
openweather	Cloud cover (cloud_cover)	grid:center (4.644 km) 48.38, grid:east (19.981 km) 51, grid:north (24.532 km) 53.51, grid:south (26.414 km) 45.19, grid:west (30.0 km) 44.58
openweather	Relative humidity (humidity)	grid:center (4.644 km) 83.62, grid:east (19.981 km) 87.14, grid:north (24.532 km) 83.53, grid:south (26.414 km) 83.12, grid:west (30.0 km) 83.06
openweather	Precipitation rate (precipitation)	grid:center (4.644 km) 0.3421, grid:east (19.981 km) 0.1251, grid:north (24.532 km) 0.1315, grid:south (26.414 km) 0.5026, grid:west (30.0 km) 0.1325
openweather	Surface pressure (pressure)	grid:center (4.644 km) 1007, grid:east (19.981 km) 1007, grid:north (24.532 km) 1007, grid:south (26.414 km) 1007, grid:west (30.0 km) 1007
openweather	Air temperature (temperature)	grid:center (4.644 km) 29.06, grid:east (19.981 km) 28.69, grid:north (24.532 km) 28.82, grid:south (26.414 km) 28.84, grid:west (30.0 km) 28.88
openweather	Wind direction (wind_direction)	grid:center (4.644 km) 146, grid:east (19.981 km) 154.1, grid:north (24.532 km) 136.6, grid:south (26.414 km) 159.9, grid:west (30.0 km) 145.9
openweather	Wind speed (wind_speed)	grid:center (4.644 km) 2.512, grid:east (19.981 km) 3.528, grid:north (24.532 km) 2.378, grid:south (26.414 km) 1.703, grid:west (30.0 km) 1.767
purpleair	PM2.5 (pm25)	222601 (2.006 km) 13.55, 229853 (2.189 km) 13.28, 308976 (2.99 km) 15.82, 133994 (3.421 km) 4.058, 186303 (3.504 km) 10.73, 235427 (3.982 km) 15.42, 127451 (4.154 km) 12.49, 27821 (5.042 km) 1.082, +62 more
tceq	Carbon monoxide (co)	482011035 (7.017 km) 0.1143, 482011052 (8.428 km) 0.3712, 482011039 (21.358 km) 0.1342
tceq	Nitrogen dioxide (no2)	482011035 (7.017 km) 6.726, 482011052 (8.428 km) 15.07, 482010416 (9.465 km) 3.749, 482010024 (14.701 km) 0.4511, 482011066 (17.679 km) 3.863, 482010047 (18.045 km) 6.611, 482010026 (18.937 km) 6.152, 482010055 (19.314 km) 2.776, +5 more
tceq	Sulfur dioxide (so2)	482011035 (7.017 km) -0.3771, 482011039 (21.358 km) 0.2466, 482010051 (22.104 km) 0.2479


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
40 km
30 km
20 km
10 km
asos (9)
openaq (103)
purpleair (70)
tceq (14)
anomaly
Latitude (degrees north)
Longitude (degrees east)
30.2
30.1
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.8
−95.6
−95.4
−95.2
−95.0


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 6.17 m/s
noaa_gfs
longest arrow = 4.52 m/s
openweather
longest arrow = 3.44 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

NO2 levels were also elevated during the same period.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

High relative humidity levels exceeding 75% during the morning hours facilitated hygroscopic growth of fine particulate matter, increasing total measured PM2.5 mass concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

A coarse-particle or mixed aerosol event was a plausible contributor because PM10 rose strongly at the same time as PM2.5, while low ozone and the absence of strong satellite SO2 enhancement do not support a purely photochemical sulfate episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

Coincident elevations in PM10 and carbon monoxide concentrations between 12:00 UTC and 18:00 UTC on June 18 indicate a primary urban emission contribution to the PM2.5 spike.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

A major regional combustion or sulfur-rich industrial plume is not strongly supported because the anomaly coincided with low ozone near 0.013 ppm, only moderate co-pollutant levels of about 0.2 ppm CO and 11.1 ppb NO_2, and no clear satellite sulfur enhancement, even though elevated PM_{10} to 68 ug/m3 indicates some possible coarse-particle contribution.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

Regional or local transport from the Gulf and the Houston Ship Channel industrial corridor was a plausible contributor because winds near the anomaly were southerly to south-southwesterly, which can advect marine aerosol and industrial particulate matter toward central Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

A shallow morning planetary boundary layer on June 18, 2026, restricted vertical dispersion and trapped particulate pollution near the ground level.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

A shallow planetary boundary layer and light surface winds limited vertical atmospheric mixing, trapping locally emitted fine particulate matter near the ground.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

High morning relative humidity levels above eighty percent enhanced hygroscopic growth of atmospheric aerosols, increasing total measured PM2.5 mass concentration.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Southerly to southeasterly surface wind flow transported and concentrated local urban and industrial particulate emissions into the affected area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

The anomaly occurred on June 18th, 2026, around 13:00:19+00:00 (0.3 min away) and was detected at a location approximately 2.006 km from the query point.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

Houston PM2.5 was elevated across both openaq and purpleair networks near 2026-06-18 13:00 UTC, indicating a real multi-sensor particulate episode rather than an isolated monitor artifact.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

Relative humidity levels above 76% in Houston during the morning of June 18, 2026, promoted hygroscopic aerosol growth that intensified PM2.5 concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The anomaly is a result of a wildfire or agricultural burning in the surrounding areas, releasing large amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

Houston meteorology likely contributed to PM2.5 accumulation because the anomaly occurred with very high near-surface humidity, light winds, and a relatively shallow morning boundary layer that would reduce dispersion and enhance hygroscopic particle growth.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The anomaly was likely caused by a combination of factors, including weather patterns and local emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Houston aerosol composition during the episode was most consistent with a mixed urban particle event with some coarse-mode contribution rather than a dominant major combustion or sulfur-rich plume, because PM10 increased while ozone remained modest and satellite CO and SO2 showed no strong enhancement.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The nearest openaq PM10 measurement at the same location and time was 68.0 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

The purpleair PM2.5 measurement nearest in time to 2026-06-18T13:00:00+00:00 and within 2.006 km of (29.769, -95.318) was 39.8 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO_2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

On June 18, 2026, at 13:00 UTC, surface air quality monitors in Houston recorded a PM2.5 peak of 42.0 $\mu g/m^3$ alongside elevated PM10 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

A shallow planetary boundary layer below 310 meters during the morning of June 18, 2026, suppressed vertical atmospheric mixing and trapped local combustion emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The openaq PM2.5 measurement at 2026-06-18T13:00:00+00:00 at (29.769, -95.318) was 42.0 ug/m3, compared with an expected value of 12.866257668711656 ug/m3, corresponding to a z-score of 6.630573005685529 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

On June 18, 2026, at 13:00 UTC, OpenAQ recorded a severe PM2.5 anomaly reaching 42.0 ug/m3 in Houston, Texas at location (29.769, -95.318) against an expected baseline of 12.87 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

At 13:00 UTC on June 18, 2026, PurpleAir sensors within 2.1 km of the anomaly site recorded elevated PM2.5 concentrations reaching 39.8 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

The PM2.5 levels are elevated at 39.8 ug/m3, which is higher than the mean of 15.5881 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

At 13:00 UTC on June 18, 2026, OpenAQ PM10 measurements at the anomaly location peaked at 68.0 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

A Houston-area accumulation episode is strongly supported because PM2.5 rose sharply and broadly across independent surface networks at the anomaly time, with openaq PM2.5 reaching 42.0 ug/m3 and purpleair PM2.5 reaching 39.8 ug/m3 near the site while 6-hour mean PM2.5 increased to about 30 ug/m3 in both networks on 2026-06-18 12Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas on June 18th was characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

Houston meteorology near 2026-06-18 13:00 UTC featured very high humidity, light southerly flow, and only moderate boundary-layer depth, conditions that favor accumulation and hygroscopic enhancement of particulate matter.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

Meteorological conditions support inefficient dispersion and moisture-enhanced particle mass because near the anomaly the air was very humid, winds were light to modest, and modeled boundary-layer depth was still relatively low for daytime, with about 94 percent relative humidity at the nearest ASOS observation, about 2.2 to 2.6 m/s surface wind near the site, and GFS planetary boundary-layer height near 417 m at 12Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.